Ashutosh Sharma

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Education

Aalto University | University of South-Eastern Norway | Helsinki, Finland | Horten, Norway
Budapest University of Technology and Economics | Budapest, Hungary
Joint International Master of Science (Technology), Smart Systems Integrated Solutions (SSIs) Aug 2023 – Aug 2025
4.28/5
Coursework: System Level (Digital) Design, Virtual Instrumentation, Materials and Microsystem Integration, Design for Reliability, Sustainable Electronics, Design and Analysis of MEMS, Multiphysics Modelling, IC and MEMS, Microfabrication, Measurement & Characterization.

Indian Institute of Information Technology Design and Manufacturing (IIITD&M), Kurnool Andhra Pradesh, India
Bachelor of Technology (Honors), Electronics and Communication Engineering June 2021
9.08/10
Coursework: Analog Circuits, Engineering Electromagnetics, Signals and Systems, Digital Logic Design, Control Systems, Digital Signal Processing, Digital IC Design, Solid State Devices, Embedded System Design, Circuits for Electronic System Design.

Experience

Tundra Drone Technologies AS Kautokeino, Norway
Embedded Electronics Engineer Dec 2025 – Dec 2026

- Designed digital, analog, and power electronics for drone payload systems including multi-layer PCBs, switching and linear power supplies, motor and LED driver stages with protection circuits (reverse polarity, TVS/ESD, current limiting).
- Designed and optimized multi-layer PCBs in KiCad: schematic capture, component selection, routing, DRC/DFM verification, and manufacturing validation; reduced fabrication cost and lead time through design-rule and panelization optimization.
- Brought prototype hardware from concept to functional boards: performed board bring-up, hardware debugging, and validation using oscilloscopes, electronic loads, and bench power supplies; diagnosed and resolved power, thermal, and signal-integrity issues.
- Integrated sensors, actuators, and communication interfaces (SPI, I²C, UART, CAN, 1-Wire, RS-485) onto airborne platforms, resolving timing, EMC, thermal, and system-level reliability challenges.
- Led prototyping, production validation, and deployment of hardware modules, collaborating across mechanical and manufacturing teams.
- Prepared schematics, BOMs, assembly drawings, and production release documentation; coordinated with fabricators and assembly partners through design review, rework, and batch production.
- Developed bare-metal firmware (STM32, PIC16/18, RP2040) for real-time control, including interrupt-driven communication, PWM, ADC, and quaternion-based stabilization.

GE HealthCare and USN Norway Horten, Norway
Master Thesis Feb 2025 – Jun 2025

- Designing and optimizing comparator for SAR ADC in 3D Medical Ultrasound, utilizing Cadence tools and UMC180nm PDK.
- Gaining solid experience with cadence tools and analog circuit design.

SINTEF DIGITAL Oslo, Norway
Summer Researcher, Silicon Sensor Technology Group Jun 2024 – Aug 2024

- Automated a chemical deposition process of molecular layer deposition by programming the movements; generated through a custom desktop software built using Python and PyQt.
- Experimented with various compounds for Raman Spectroscopy for certain applications.

Mindtree Bengaluru, India
Engineer, Software Development Jun 2021 – Apr 2022

- Building Windows desktop applications as per the requirements using C++ and MFC, developed practices to work with git version control systems.
- Developing macOS-based application using objective C.

VLSI System Design Corporation Bengaluru, India
Analog IP Research Intern Jul 2020 – Sep 2020

- Designed a ***10-bit potentiometric digital-to-analog converter*** as per the specifications given by VSD Corporation using open-source EDA tools and osu180nm PDK.
- Gained experience with tools required during the process such as eSim, ngspice, and Magic
- Automated post-layout characterization and analysis using R language.

Projects

- *Design of an Accelerometer* Feb 2024 – Dec 2024
  - Utilized *COMSOL* simulations and *ANSYS workbench* to design an accelerometer and analyze the spring constant and resonant frequency, enabling a thorough comparison with analytical results.
  - Designed an analog readout circuit using *Cadence Virtuoso* and *AMS 0.65µm PDK*. Converted the signal for digital processing using VCO design.
  - Utilised *ANSYS Twinbuilder* to develop mixed signal design test environment to simulate the developed ROM model with analog and digital circuit components.

- *Design of an Operational Amplifier* Nov 2024 - Dec 2024
  - Degined an operational amplifier for 1000x closed loop gain for a MEMS accelerometer design to convert the capacitance(fF) into voltage(mV) to the range of one volts.
  - Performed DC, Transient, AC, Stability Analysis, PVT and Monte Carlo to verify the schematic of the circuit.
  - Gained insights into topologies of amplifiers, cadence design suite.
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- *Klein | a Physiological Parameter Monitoring System* Jan 2021 – May 2021
  - Developed a PCB for the device which monitors and transmits blood oxygen saturation (SpO_2), ECG waveform, body temperature, and heart rate.
  - Included the feature to continuously display the current date, time, day, and month along with the vital body parameters.
  - Exploited the features of the microcontroller, such as ADC, interrupts, scheduling, to achieve the required functionality.

- *Audio and Image Compression* May 2020– June 2020
  - Extensively researched integer DCT and related transforms for efficient design.
  - Implemented audio compressing techniques - integer DCT transformation technique using MATLAB.
  - Implemented image compression using jpeg image compression using MATLAB.

Skills & Interests

Technical: C++, Python, shell scripting, VHDL, C
Software & Tools: Kicad, STMCube, MPLAB, Cadence, ANSYS workbench, APDL, ModelSim, COMSOL Multiphysics, LabVIEW, MATLAB, eSim, ngspice, Magic, SolidWorks, LTspice, Git, VS Code.
Interests: Embedded Systems, MEMS, Analog Design, Microfabrication, Software Development.
Hardware: STM32, PIC16, CYCLONE III, TIVA (TM4C123GH6PM), ATmega-16, ESP32.

Honours & Awards

- *Completed* **SSIs - Joint International Masters** *with* **Distinction** Aug 2025
- ***1st Runner Up*** in **The Student Expo** at the **Sensor Decade 2024, Oslo** for project **RoMedica.** Jun 2024
- ***Awarded*** the **Erasmus Mundus Scholarship** to pursue EMJMD SSIs. Apr 2023
- ***Top 5%*** among ***~46K*** students appeared in ***the*** **GATE Exam, 2023** - Electronics and Communication Engineering. Mar 2023
- ***Received*** **Roll of Honor Gold Medal** from the Electronics and Communication Department of the Indian Institute of Information Technology Design and Manufacturing (**IIITD&M**), Kurnool. Sep 2021

- ***Selected to present*** the open-source Analog IP – "10-bit potentiometric DAC 3.3V analog voltage, 1.8V digital voltage and 1 off-chip external voltage reference" at the **VSDOpen2020 conference**. Oct 2020
- Selected for ***participation*** in **MIT-India Initiative** in housing and energy track held at Welingkar School Mumbai, India. Jan 2020
- ***One*** of the 747 teams selected in the semifinals of the **Indian Innovation Challenge Design Contest (IICDC)** out of 31,000 teams conducted by DST and **Texas Instruments**, India. Jan 2020